

**Mini- Assessment - Capstone Lab - 27 Feb 2026****End-to-End CX Training Request Automation (Event-Driven Notifications)****Scenario**

Build a complete system to manage Employee Training Requests. The platform must:

- Capture requests
 - Enforce UI rules
 - Validate data
 - Automate approvals
 - Calculate risk
 - Send notifications using events
 - Maintain data integrity
 - Use both client-side and server-side logic appropriately
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PRE-SETUP**Step 0 — Create Update Set**

Create and make current an Update Set with the following details:

Name: CAPSTONE - CX Training Automation - Full Stack

Platform Build and Client-Side Logic**TASK 1 — Create Main Table**

Create a new table with these properties:

Label: CX Training Request

Name: u_cx_training_request

Extends table: Task

Required Fields

Request Details

- u_requested_for — Reference to User
- u_training_type — Choice
 - Technical
 - Functional

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- Compliance
- u_urgency — Choice
 - Low
 - Medium
 - High
 - Critical
- u_cost — Integer
- u_reason — String
- u_approval_required — True/False
- u_manager — Reference to User
- u_approval_status — Choice
 - Draft
 - Pending
 - Approved
 - Rejected

Automation Fields

- u_auto_score — Integer
 - u_validation_notes — String
 - u_processed — True/False
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TASK 2 — Create Sample Records (Manual)

Manually create at least five records with different combinations of:

- Urgency levels
 - Cost values
 - Training types
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TASK 3 — Dictionary Override

Goal: Override inherited Task field behavior.

Field: Short Description

Requirements:

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- Mandatory only for CX Training Request table
- Help Text: "Provide clear request summary"

Steps:

1. Navigate to System Definition → Dictionary
 2. Locate short_description for your table
 3. Create a dictionary override if inherited
 4. Set Mandatory to true
 5. Add help text
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TASK 4 — UI Policies (No Scripts)

UI Policy 1 — High Urgency Requires Reason

Condition: u_urgency is High OR Critical

Actions:

- u_reason mandatory = true
- Reverse if false = true

UI Policy 2 — Functional Training Hides Cost

Condition: u_training_type = Functional

Actions:

- u_cost visible = false
 - Reverse if false = true
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TASK 5 — Client Script (onLoad)

Goal: Default initialization and user guidance.

Requirements:

1. If urgency is empty, set it to Medium
2. Show info message:
"Please fill all mandatory fields before submission"
3. If approval status is Approved, make fields read-only

Must use:

- g_form.setValue()
- g_form.addInfoMessage()

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- `g_form.setReadOnly()`
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TASK 6 — Client Script (onChange: Cost)

Field: `u_cost`

Rules:

1. If `cost < 0`
 - Clear value
 - Show error message
2. If `cost > 20000`
 - Show warning message
3. If `cost > 50000`
 - Show alert popup

Must use:

- `g_form.getValue()`
 - `g_form.clearValue()`
 - `g_form.showFieldMsg()`
 - `alert()`
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TASK 7 — Client Script (onChange: Requested For)

Goal: Auto-populate manager field.

When `u_requested_for` changes:

- Set `u_manager` to the selected user's manager

Hint: Use reference field retrieval via `g_form`

Server-Side Automation

TASK 8 — Data Generation Script

Create a background script that programmatically inserts 10 to 20 records.

Requirements:

- Random urgency
- Random cost

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- Requested_for = current user
- Approval status = Draft

Must use:

- GlideRecord
 - initialize()
 - insert()
 - gs.getUserID()
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TASK 9 — Auto-Validation Business Rule

Type: Before Insert and Before Update

Logic:

If either condition is true:

- Cost > 15000
- Urgency = Critical

Then:

- Set approval_required = true
- Add info message: "Manager approval required"

Must use:

- gs.addInfoMessage()
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TASK 10 — Manager Auto-Assignment

Type: Before Insert and Before Update

Logic:

If approval_required is true AND manager is empty:

- Assign manager from Requested_for.manager

Use GlideRecord reference access.

TASK 11 — Risk Scoring Logic

Calculate u_auto_score using the following rules:

- Cost > 20000 adds 50 points

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- Urgency Critical adds 40 points
- Training Type Compliance adds 20 points

Store the total score in `u_auto_score`.

TASK 12 — Validation Notes Generator

Populate `u_validation_notes` with textual reasons such as:

- High cost request
- Critical urgency
- Compliance training

Combine multiple reasons if applicable.

TASK 13 — Approval Status Automation

Set approval status based on score:

- Score ≥ 80 $\rightarrow$ Pending
 - Score < 80 $\rightarrow$ Approved
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TASK 14 — Process Flag

Type: After Insert Business Rule

Action:

- Set `u_processed` = true
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TASK 15 — Data Cleanup Script

Create a background script that deletes invalid records where:

- Cost < 0
OR
- Requested_for is empty

Log details before deleting each record using `gs.info()`.

TASK 16 — Reporting Script

Generate a system summary using a background script:

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- Total requests
- Pending approvals
- Approved requests
- Average cost
- Highest cost request
- Count of Critical urgency

Print results using `gs.info()`.

TASK 17 - Event-Driven Notifications**TASK 17A — Register Custom Events**

Navigate to System Policy → Events → Registry and create:

Event 1

Name: `u.cx.training.pending`

Table: `u_cx_training_request`

Description: Training request moved to Pending approval

Event 2

Name: `u.cx.training.autoapproved`

Table: `u_cx_training_request`

Description: Training request auto-approved

TASK 17B — Event Queue Business Rule

Create a Business Rule to fire events when approval status changes.

Name: BR - Queue CX Training Events

Table: `u_cx_training_request`

When: After

Insert: Optional

Update: Enabled

Logic:

If approval status changes to Pending:

- Queue event `u.cx.training.pending`

If approval status changes to Approved:

- Queue event `u.cx.training.autoapproved`

Parameters to pass:

- `parm1`: Request Number

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- parm2: Requested For display name

Must use:

- changesTo() condition
 - gs.eventQueue()
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TASK 17C — Email Notification (Pending Approval)

Create a notification triggered by event.

Name: CX Training - Pending Approval

Table: u_cx_training_request

When to send: Event is fired

Event: u.cx.training.pending

Recipients:

- Manager (u_manager field)

Email content must include:

- Request number
- Requested for
- Cost
- Urgency
- Reason
- Link to record

Verify using Outbound Email logs.

TASK 17D — Email Notification (Auto Approved)

Name: CX Training - Auto Approved

Event: u.cx.training.autoapproved

Recipients:

- Requested For
- Request creator (optional)

Content:

- Approval message
- Request details
- Explanation of auto-approval

BONUS CHALLENGE

Automatically reject requests if:

Cost > 50000 AND Urgency = Low

Actions:

- Set approval_status = Rejected
 - Add validation note explaining rejection
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FINAL SUBMISSION REQUIREMENTS

Screenshots Required

1. Update set marked current
 2. Table structure and fields
 3. Dictionary override proof
 4. UI Policies configuration
 5. Client Scripts list
 6. Business Rules list
 7. Sample records
 8. Event Registry entries
 9. Notification configuration
 10. Outbound email proof
 11. Event queue Business Rule
 12. Reporting script logs
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Written Answers

Provide short explanations for:

1. Difference between Client Script and Business Rule
2. Difference between UI Policy and Client Script
3. GlideSystem vs GlideRecord
4. Why dictionary override is used
5. When to use server-side vs client-side logic.

